

RUI SOUSA

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EDUCATION

Northwestern University, Evanston, IL

2021-2027 (Expected)

PhD Student, Economics Department

- Master of Arts, 2022

- Relevant coursework: Graduate Econometrics, Macro, Micro, Asset Pricing & Corporate Finance (Kellogg Business School)

Nova School of Business and Economics, Lisbon, Portugal

2016-2019

MSc Economics, GPA 19/20 (A) - Novo Banco Master's Scholarship

- Exchange Program at University of Copenhagen

University of Porto, Porto, Portugal

2013-2016

BSc Economics, GPA: 18/20 (A)

RESEARCH ACTIVITY

The Aggregate Green Elasticity of Substitution (**Draft**)

- Using a novel semi-structural model, provide the first quantification of the US-wide elasticity of substitution between clean and polluting energy sources – a key parameter governing energy prices pass-through, and the macroeconomic costs of decarbonization
- After deriving a new equilibrium relationship between energy consumption and observable prices, exploit U.S. states' heterogeneous energy mixes to implement a shift-share instrumental variables strategy, estimating an aggregate elasticity of 0.5
- Construct an analytical mapping between sectoral elasticities and their aggregate counterpart. Using 2022 data to calibrate the model, find that energy end-users electrify more easily than the electricity sector shifts to green sources, recovering elasticities of 0.8 and 0.52, respectively

Selected Work in Progress:

- *The Adaptation-Mitigation Trade-off*
 - Quantify the optimal mix between adaptation and mitigation investment. Study the welfare implications of adaptation investment when mitigation technologies exhibit learning-by-doing, highlighting coordination failures in climate policy
- *Climate Change Adaptation and the Sovereign Debt Overhang*
 - Document the rising frequency and severity of climate disasters across emissions pathways. Calibrate a sovereign default model with disasters to study their implications to default probabilities and investment adaptation in emerging markets

PROFESSIONAL EXPERIENCE

London School of Economics, London, United Kingdom

Sep 2020 - Sep 2021

Research Assistant, Centre for Macroeconomics. Advisors: Profs. Benjamin Moll & Ricardo Reis

- Solved continuous-time heterogeneous agent models numerically
- Modeled rare inflation events implementing threshold selection algorithms and maximum likelihood estimation
- Analyzed consumer and professional inflation expectations microdata

European Central Bank, Frankfurt am Main, Germany

Jun 2018 - Sep 2021

Analyst, Crisis Management - Quantitative Cluster

- Contributed to early warning models processing 1M+ daily observations from global media database and implementing natural language processing
- Analyzed liquidity provisions during stress events and the impact of banking policy on profitability
- Contributed to the impact assessment of Basel III on MREL requirements

Trainee, Stress-test Modelling

- Implemented causal inference methods (RDD, DiD) on large supervisory banking panel datasets to identify the impact of stress-tests and capital requirements on risk-taking and profitability
- Contributed to credit risk assessment tool using historical loan portfolio data

TOOLS & SKILLS

- **Programming:** Python, R, Matlab
- **Computing:** SLURM (HPC), Git, $\LaTeX$
- **Datasets:** Compustat, CRSP, CES, PSID, SPF, SCE, MER, SEDS, CMIP6
- **Languages:** English, Portuguese (fluent); French, Spanish (conversational)
- **Time-series:** State-space & nonlinear models, time-varying volatility, Bayesian inference, forecasting
- **Microeconometrics:** Shift-share IV, DiD, RDD, statistical learning
- **Modelling:** Dynare, Sequence-space Jacobian toolbox, Deep Equilibrium Networks